

Regression & 1D CNN Result Comparison with AEC, SMI

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1. What's discussed in the last meetings

- SMI 데이터 확립 → TAMA, BMI, Height, Weight 등의 데이터로 확인 가능
- AEC data Resampling → 256 point로 정규화
 - 이후 주요 feature 추출 (mean, std, range, skewness 등 65개 feature)
- 이전 연구와 동일하게 비교 분석

Model	Feature 구성	Feature_num
clinic_only	PatientAge, PatientSex, BMI	3
aec_only	aec1 ~ aec256 (원시 AEC 시계열 통계)	256
aec_feature_only	aec_feature_filtered 시트 전체 사용	65
aec_feature_selected	AEC 피쳐 fold별 선택 ($ r < 0.8$, $VIF < 10$)	가변
clinic_aec_feature	Age/Sex/BMI (고정) + AEC 피쳐 fold별 선택	가변

2. Research Summary

- IMATA 값을 반영하여 SMI 데이터 재확인
 - $SMI = (TAMA - IMATA) / Height^2$
- SMI 기준 논문 5개 확인 후 남녀 SMI threshold 결정
 - ‘한국 기준 하위 5%’로 설정
- 모델은 Logistic Regression과 ResNet1D만 사용 (Linear 제외)
- Clinic [Age, Sex, BMI] vs Clinic + AEC 256 points vs Clinic + AEC +Scan[kVp, 제조사]

연구 프로토타입 정의

Research Overview – Main Study

- 연구 목적

- Opportunistic CT Biomarker 기반 → 5년 내 vertebral compression fracture 예측
- AEC의 incremental value 검증

- 데이터

- Baseline APCT + follow-up imaging 보유 환자
- 주요 변수: Age, Sex, AEC, Muscle, Bone Density 등

- 모델 구조

- Model A: Age + Sex
- Model B: + Bone
- Model C: + Muscle
- Model D: + AEC (핵심 비교)

Research Overview – Pilot Feasibility Study (현재 진행 연구)

- 연구 목적

- AEC가 단순 보조 변수인지 vs 독립적인 biomarker인지 검증
→ 본 연구 모델 포함 가능성 평가

- 핵심 질문

- AEC는 age/sex 이상 정보를 가지는가?
- Muscle/Bone과 독립적 연관성이 존재하는가?
- Scanner/Center 보정 후에도 유지되는가?

Research Overview – Pilot Feasibility Study (현재 진행 연구)

- 분석 전략

- 분포 분석 (AEC by 센터, 성별, 연령) – Complete
- 회귀 분석 (muscle area / attenuation / bone density ~ AEC + age + sex)
- 고급 분석
 - Multivariable (center, scanner 보정)
 - Spline 분석 → 비선형 관계 확인
 - Center-stratified 분석

- 결과물

- AEC distribution plot
- AEC vs biomarker scatter
- Regression Table
- Stratified Result Figure 등

데이터 정리

3. Detailed progress

- Data – 강남
 - Clinic Data 수 : 3,672 → 중복 PatientID 제거 : 2,025 → TAMA outlier(0, 1), BMI 제거 : 1,642
 - Axial Data 수 : 1,642 → Fixed Tube Current Value 제거 : 1,372

총 환자 수
1,372명

남성
514명 (37.4%)

여성
858명 (62.6%)

SMI 분포 $SMI = (TAMA - IMATA) / Height(m)^2$

성별	N	mean	std	P25	median
Female	858	38.27	5.95	34.58	37.67
Male	514	48.38	8.54	43.27	48.42

TAMA 분포

성별	N	mean	std	P25	median
Female	858	104.98	15.28	95	103
Male	514	148.23	26.35	132	148

3. Detailed progress

- 데이터 분포 확인

구분	Mean	Std	Min	p25	Median	p75	max
n slices	168.6	32.0	44	151	165	184	460
z range (mm)	469.6	52.6	215.0	441.0	467.5	497.5	918.0

구분	성별	Mean	Std	Min	Median	Max
n slices (개수)	F	168.61	32.05	50.0	165.0	460.0
	M	168.65	32.04	44.0	166.0	405.0
z range (mm) (범위)	F	470.97	51.47	234.0	468.0	918.0
	M	467.58	54.34	215.0	465.0	808.0

3. Detailed progress

- Train / Test Dataset Split & Distribution

Split	Sex	n	Normal	Normal %	Sarco	Sarco %
Train	M	402	332	82.6%	70	17.4%
	F	695	645	92.8%	50	7.2%
	All	1097	977	89.1%	120	10.9%
Test	M	112	95	84.8%	17	15.2%
	F	163	150	92.0%	13	8.0%
	All	275	245	89.1%	30	10.9%

3. Detailed progress

- Train / Test Dataset Split & Distribution – Age

Split	Sex	n	Mean ± Std	Min	Median	Max
Train	M	402	59.81 ± 12.51	18.00	60.00	89.00
	F	695	55.36 ± 12.15	11.00	55.00	91.00
	All	1097	56.99 ± 12.47	11.00	58.00	91.00
Test	M	112	59.05 ± 12.52	23.00	59.50	84.00
	F	163	56.52 ± 12.29	22.00	56.00	87.00
	All	275	57.55 ± 12.45	22.00	58.00	87.00

3. Detailed progress

- Train / Test Dataset Split & Distribution – BMI

Split	Sex	n	Mean $\pm$ Std	Min	Median	Max
Train	M	402	24.22 $\pm$ 3.26	14.48	24.19	36.76
	F	695	23.09 $\pm$ 3.43	14.40	22.70	39.49
	All	1097	23.51 $\pm$ 3.41	14.40	23.30	39.49
Test	M	112	24.07 $\pm$ 3.30	16.44	24.16	35.20
	F	163	22.99 $\pm$ 3.19	16.06	22.83	34.23
	All	275	23.43 $\pm$ 3.28	16.06	23.44	35.20

SMI Cutoff Setting

3. Detailed progress

- 성별에 따른 SMI Cutoff Value 분석

연구 논문 (저자)	대상 국가	대상군 특성	평균 나이 (Mean±SD)	나이 범위 (Range)
Prado et al. (2009)	캐나다	전이성 유방암 환자	54.8 ± 10.4세	37~80세
Yoon et al. (2021)	한국	건강한 간 기증자	31.5 ± 8.3세	20~59세
JSH (2016)	일본	만성 간 질환 환자	71세 (중앙값)	19~89세
Carey et al. (2017)	북미	간 이식 대기 환자	58세 (중앙값)	(IQR: 51~62세)
Martin et al. (2013)	캐나다	폐암 및 소화기암 환자	남성: 64.7세 / 여성: 64.8세	(기술되지 않음)
현재 연구 데이터			남성 59.7 ± 12.5세 여성 55.6 ± 12.2세	11~91세

3. Detailed progress

- 성별에 따른 SMI Cutoff Value 분석

연구 논문 (연도)	대상 국가	남성 Cutoff	여성 Cutoff	남성 비율 (강남)	여성 비율 (강남)
Prado et al. (2009)	캐나다	52.4	38.5	69.3% (356/514)	57.2% (487/858)
Martin et al. (2013)	캐나다	53.0	41.0	71.8% (369/514)	73.3% (627/858)
Carey et al. (2017)	미국/캐나다	50.0	39.0	58.2% (299/514)	60.8% (518/858)
JSH (2016)	일본	42.0	38.0	20.0% (103/514)	52.9% (449/858)
Yoon et al. (2021)	한국	39.33 (-2 STD)	27.7 (-2 STD)	12.3% (63/514)	3.6% (22/858)
		40.96 (하위 5%)	30.6 (하위 5%)	17.3% (87/514)	8.4% (63/858)

연구 결과

3. Detailed progress

- CV Summary – Model 1 [Clinic]

- Logistic Regression

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.8329	0.3651	0.1963	0.6864	0.3670
2	0.8095	0.3370	0.1612	0.7591	0.3765
3	0.7784	0.3403	0.2040	0.6804	0.3137
4	0.7233	0.2329	0.1830	0.7306	0.3059
5	0.8500	0.5234	0.1726	0.7489	0.3956
Mean	0.7988	0.3598	0.1834	0.7211	0.3517
±Std	0.0448	0.0936	0.0155	0.0322	0.0355

- ResNet1D

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.8703	0.4683	0.2194	0.6409	0.3577
2	0.8444	0.3939	0.3022	0.4955	0.2930
3	0.7806	0.3370	0.0961	0.8584	0.1622
4	0.8415	0.4100	0.1840	0.8265	0.4062
5	0.8517	0.3597	0.1172	0.8265	0.4412
Mean	0.8377	0.3938	0.1838	0.7296	0.3321
±Std	0.0303	0.0452	0.0740	0.1400	0.0984

3. Detailed progress

- Test Set Performance – Model 1 [Clinic]

- Overall

Model	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	0.7527	0.3338	0.1992	0.7091	0.3333
ResNet1D	0.7578	0.3240	0.3256	0.4691	0.2700

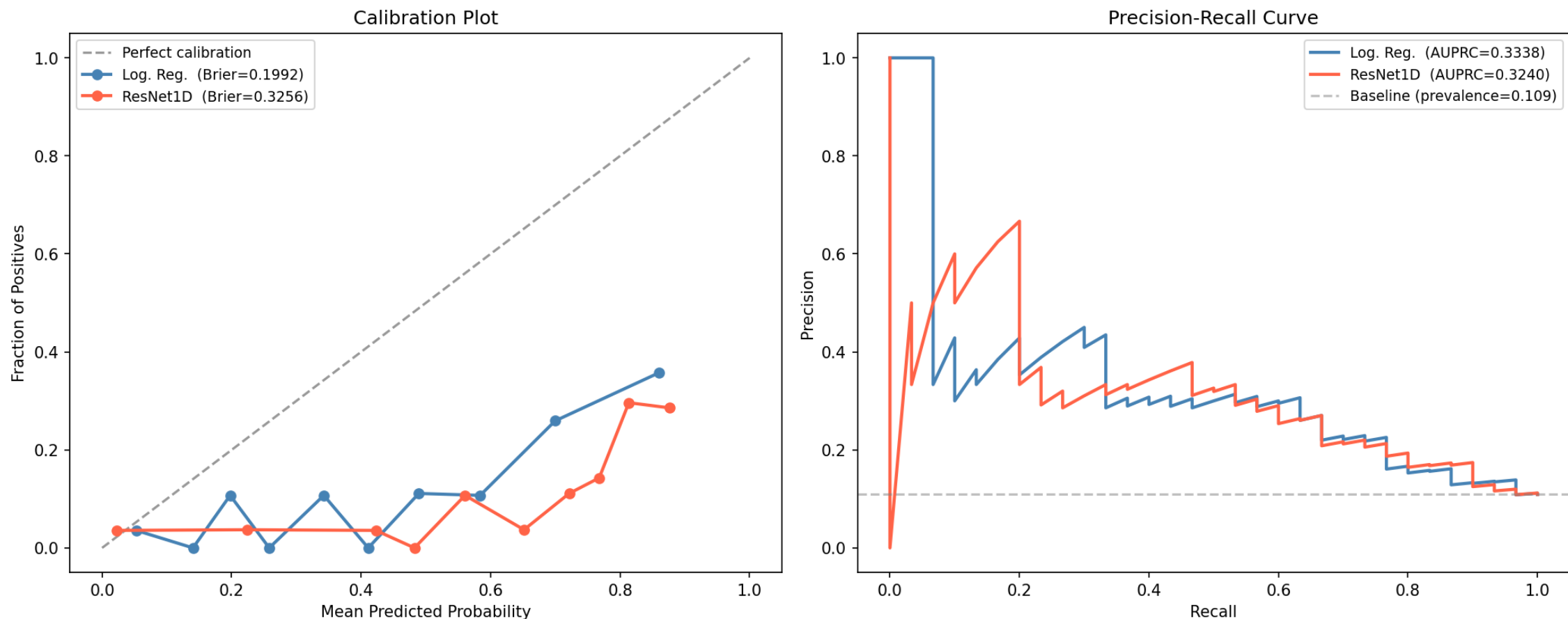
- Logistic Regression

Model	Sex	n	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log.Reg.	M	112	0.7245	0.4141	0.2790	0.5357	0.3333
	F	163	0.7682	0.2498	0.1444	0.8282	0.3333
ResNet1D	M	112	0.7152	0.3669	0.3875	0.3839	0.3030
	F	163	0.7985	0.3133	0.2830	0.5276	0.2376

3. Detailed progress

- Calibration Plot & PR Curve – Model 1 [Clinic]

Calibration & Precision-Recall Curves — Test Set



3. Detailed progress

- CV Summary – Model 2 [Clinic + AEC]

- Logistic Regression

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.5512	0.1463	0.2465	0.6636	0.2128
2	0.6482	0.1913	0.1779	0.7591	0.2740
3	0.6662	0.2269	0.1950	0.7169	0.2619
4	0.6397	0.1946	0.1923	0.7306	0.2532
5	0.6316	0.1876	0.2120	0.7169	0.2439
Mean	0.6274	0.1893	0.2047	0.7174	0.2491
±Std	0.0398	0.0257	0.0235	0.0310	0.0207

- Cross Attention

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.7651	0.3361	0.1945	0.7000	0.3400
2	0.8253	0.4046	0.1680	0.7818	0.3846
3	0.7692	0.3508	0.2285	0.6484	0.3186
4	0.6979	0.2423	0.2551	0.5799	0.2698
5	0.8825	0.5270	0.1503	0.7580	0.4301
Mean	0.7880	0.3721	0.1993	0.6936	0.3486
±Std	0.0622	0.0934	0.0384	0.0734	0.0550

3. Detailed progress

- Test Set Performance – Model 2 [Clinic + AEC]

- Overall

Model	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	0.6133	0.2317	0.2039	0.6873	0.2321
CrossAttn	0.7071	0.2936	0.1801	0.7200	0.3304

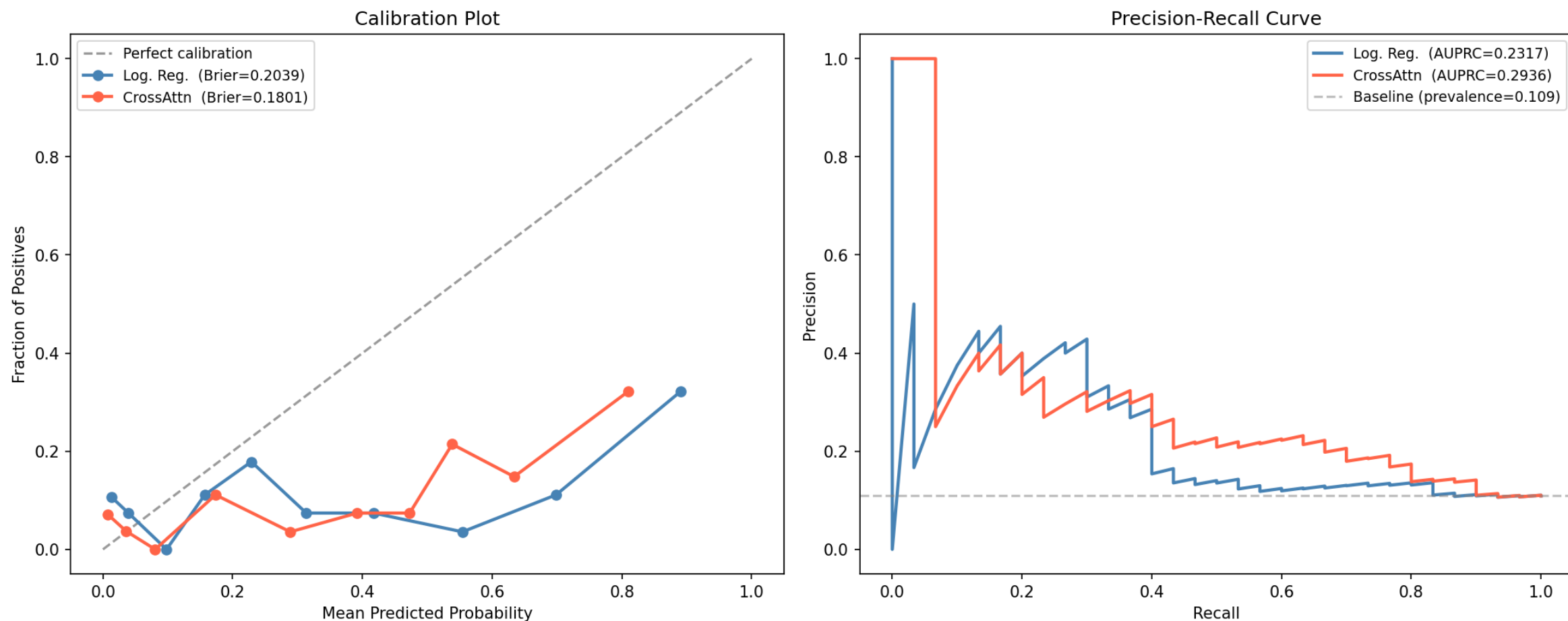
- Logistic Regression

Model	Sex	n	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	M	112	0.6495	0.3322	0.2387	0.6429	0.3103
	F	163	0.5477	0.1627	0.1801	0.7178	0.1481
CrossAttn	M	112	0.6774	0.3597	0.2495	0.5893	0.3429
	F	163	0.7092	0.2638	0.1323	0.8098	0.3111

3. Detailed progress

- Calibration Plot & PR Curve – Model 2 [Clinic + AEC]

Calibration & Precision-Recall Curves – Test Set



3. Detailed progress

- CV Summary – Model 2_2 [Clinic + AEC (Unmatching)]

- Logistic Regression

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.6144	0.1641	0.2094	0.7182	0.2250
2	0.6386	0.1758	0.2066	0.7091	0.2195
3	0.5829	0.1611	0.2081	0.6986	0.2143
4	0.7147	0.2185	0.1734	0.7808	0.3143
5	0.7160	0.2467	0.2096	0.7078	0.3043
Mean	0.6533	0.1932	0.2014	0.7229	0.2555
±Std	0.0537	0.0337	0.0141	0.0296	0.0442

- Cross Attention

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.8335	0.4235	0.1913	0.7000	0.3654
2	0.8386	0.3855	0.1378	0.8318	0.4308
3	0.7374	0.2623	0.2002	0.6941	0.3366
4	0.7682	0.2711	0.1571	0.7945	0.4000
5	0.8173	0.5119	0.2656	0.5160	0.2933
Mean	0.7990	0.3708	0.1904	0.7073	0.3652
±Std	0.0396	0.0945	0.0439	0.1095	0.0479

3. Detailed progress

- Test Set Performance – Model 2_2 [Clinic + AEC (Unmatching)]

- Overall

Model	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	0.6189	0.1600	0.2125	0.7127	0.2617
CrossAttn	0.7140	0.3629	0.1843	0.7636	0.3564

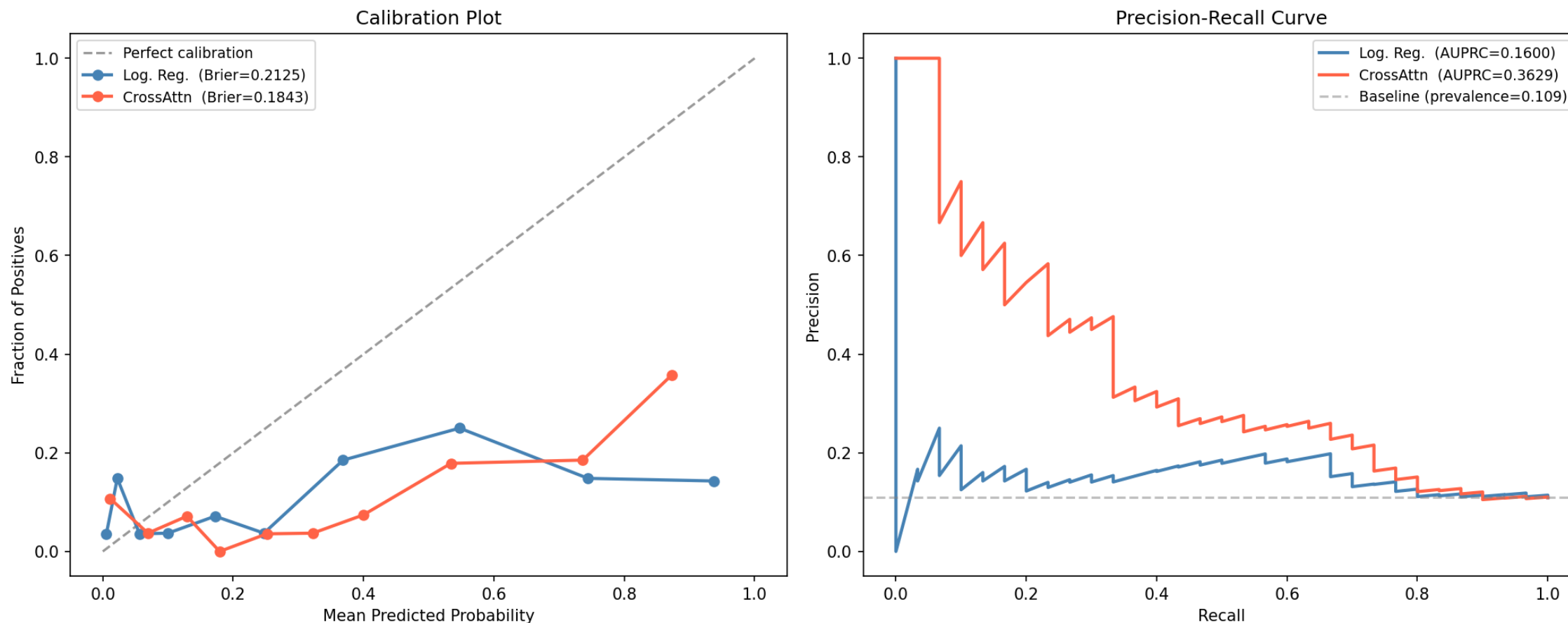
- Logistic Regression

Model	Sex	n	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	M	112	0.5895	0.2297	0.2262	0.6875	0.2857
	F	163	0.6441	0.1325	0.2031	0.7301	0.2414
CrossAttn	M	112	0.7059	0.3746	0.2677	0.5982	0.3284
	F	163	0.6882	0.3553	0.1269	0.8773	0.4118

3. Detailed progress

- Calibration Plot & PR Curve – Model 2_2 [Clinic + AEC (Unmatching)]

Calibration & Precision-Recall Curves — Test Set



3. Detailed progress

- CV Summary – Model 3 [Clinic + AEC + Scan]

- Logistic Regression

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.6350	0.2601	0.1896	0.7548	0.2609
2	0.7452	0.2260	0.1925	0.7404	0.3864
3	0.7058	0.2919	0.1691	0.7885	0.2667
4	0.5431	0.1507	0.2444	0.6587	0.2022
5	0.7177	0.2781	0.1782	0.7391	0.3077
Mean	0.6694	0.2414	0.1948	0.7363	0.2848
±Std	0.0728	0.0504	0.0262	0.0427	0.0609

- Cross Attention

Fold	AUC-ROC	AUPRC	Brier	Accuracy	F1
1	0.7986	0.3567	0.1268	0.8269	0.3793
2	0.8021	0.3632	0.1994	0.5962	0.3115
3	0.8195	0.4050	0.1189	0.8221	0.4127
4	0.7727	0.2650	0.2901	0.5337	0.3022
5	0.8025	0.3577	0.2365	0.6570	0.3364
Mean	0.7991	0.3496	0.1944	0.6872	0.3484
±Std	0.0150	0.0459	0.0651	0.1187	0.0418

3. Detailed progress

- Test Set Performance – Model 3 [Clinic + AEC + Scan]

- Overall

Model	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	0.6713	0.1958	0.1956	0.7154	0.2745
CrossAttn	0.7542	0.2410	0.1717	0.7423	0.2947

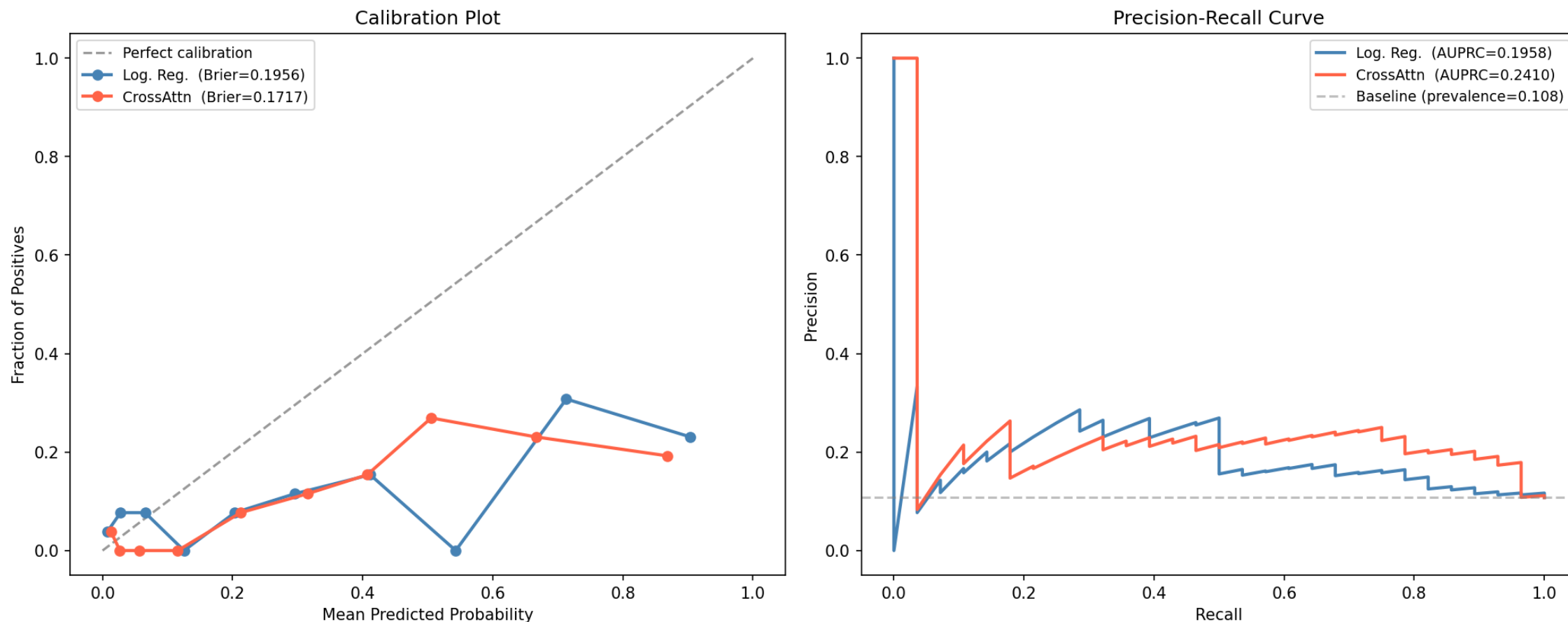
- Logistic Regression

Model	Sex	n	AUC-ROC	AUPRC	Brier	Accuracy	F1
Log. Reg.	M	98	0.6707	0.2723	0.2233	0.6939	0.3750
	F	162	0.6539	0.1399	0.1788	0.7284	0.1852
CrossAttn	M	98	0.6890	0.2373	0.2437	0.6327	0.3571
	F	162	0.7428	0.2432	0.1281	0.8086	0.2051

3. Detailed progress

- Calibration Plot & PR Curve – Model 3 [Clinic + AEC + Scan]

Calibration & Precision-Recall Curves — Test Set





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